	Marwadi University **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: FPGA Based System Design (01CT0718)	**Aim: mplement multiplexers and demultiplexers on the FPGA development board**	
Experiment No: 04	**Date:**	**Enrolment No: 92301733025**

Aim: **implement multiplexers and demultiplexers on the FPGA development board**

Software required:

Hardware required:

Theory:

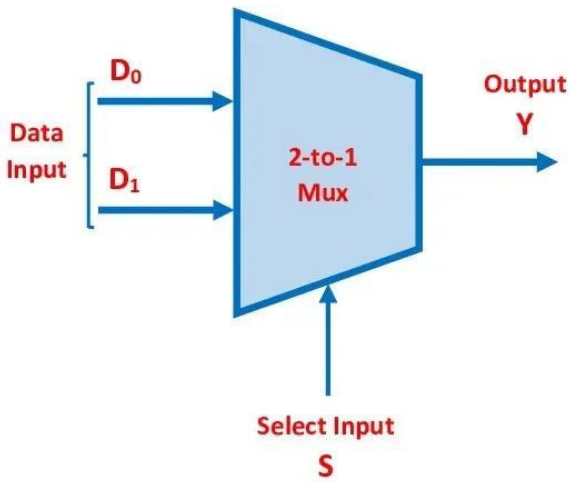
Multiplexer

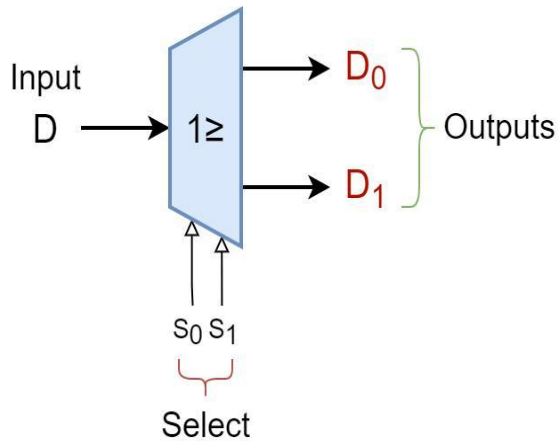
A **Multiplexer (MUX)** is a combinational circuit that selects one input from multiple inputs and sends it to a single output. The selection is controlled by **select lines**. In FPGA, multiplexers are used for data selection and routing.

Demultiplexer

A **Demultiplexer (DEMUX)** is a combinational circuit that transfers one input to one of many outputs. The output is selected using **select lines**. It is commonly used for data distribution in digital systems.

Block diagram:





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Truth Table:

MUX:

S	Output (Y)
0	I0
1	I1

DEMUX:

S	D	Y0	Y1
0	0	0	0
0	1	1	0
1	0	0	0
1	1	0	1

Boolean Expression:

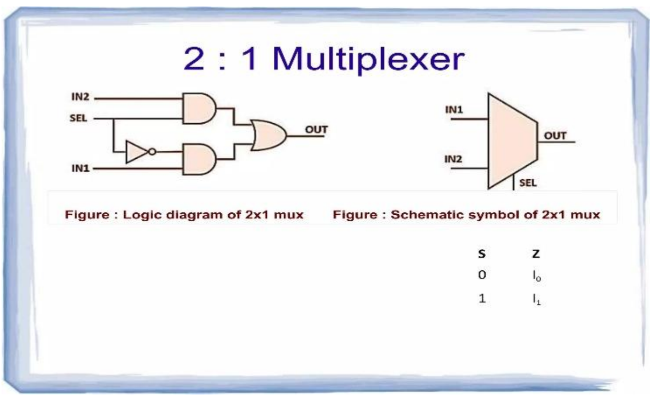
Boolean Expression (2:1 MUX)

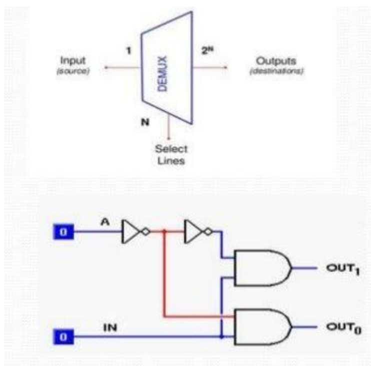
$$Y = \bar{S} I_0 + S I_1$$

Boolean Expression (1:2 DEMUX)

$$Y_0 = \bar{S} \cdot D$$
$$Y_1 = S \cdot D$$

Circuit Diagram:





TRUTH TABLE

INPUTS		OUTPUTS	
A	IN	OUT_1	OUT_0
0	0	0	0
0	1	0	1
1	0	0	0
1	1	1	0

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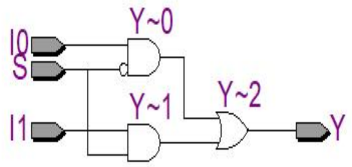
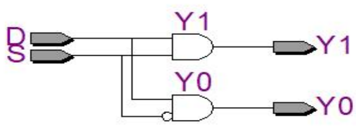
Verilog Code:

Simulation Results:

MUX



DEMUX



(FPGA Implementation)

Conclusion:

Post Lab Exercise: